

第十一章

振动与波动

1. 振动: T - 物理量 x 在某一定值附近来回变化

几个振动由 $x = A \cos(\omega t + \phi)$ 的振动的叠加

2. 谐振动: 理想模型 $m, \phi=0, f_{\text{外}}=0$



$$\begin{cases} F = -kx \\ F = m \frac{d^2x}{dt^2} \end{cases}$$

$\therefore$ 令 $\omega = \sqrt{\frac{k}{m}}$ 则 $\frac{d^2x}{dt^2} + \omega^2 x = 0 \leftarrow$ 谐振动运动方程

利用能量守恒: $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = C$

$$m v \frac{dv}{dx} + k x \frac{dx}{dt} = 0$$

$$\frac{dx}{dt} + \frac{k}{m} x = 0$$

① 求解运动方程 $\frac{d^2x}{dt^2} + \omega^2 x = 0$

解得: $x = A \cos(\omega t + \phi) \rightarrow$ 振动方程

振动曲线: 位移 x 随时间 t 变化的曲线.

② 描述谐振动的物理量

振幅 A 角频率 $\omega = 2\pi$ 秒内振动次数 (描述向运动的快慢)

ϕ : 初相位

$$0 \leq \phi < 2\pi \text{ 或 } -\pi < \phi \leq \pi$$

$$\omega = \sqrt{\frac{k}{m}}$$

③ $v = \frac{dx}{dt} = -A\omega \sin(\omega t + \phi) = A\omega \cos(\omega t + \phi + \frac{\pi}{2})$

$$a = \frac{dv}{dt} = -A\omega^2 \cos(\omega t + \phi) = A\omega^2 \cos(\omega t + \phi + \pi)$$

固有周期 $T = \frac{2\pi}{\omega}$ 固有频率 $\nu = \frac{\omega}{2\pi}$

谐振动周期和频率取决于振动系统本身的力学性质

由 $\begin{cases} \omega \\ \text{初始条件 } t=0 \text{ 时 } v_0, x_0 \text{ 共同决定} \end{cases}$

$$\begin{cases} x = A \cos(\omega t + \phi) \\ v = -A\omega \sin(\omega t + \phi) \end{cases}$$

$$t=0 \text{ 时: } \begin{cases} x_0 = A \cos \phi \Rightarrow \cos \phi = \frac{x_0}{A} \\ v_0 = -A\omega \sin \phi \Rightarrow \sin \phi = -\frac{v_0}{A\omega} \end{cases}$$

$$A = \sqrt{x_0^2 + \frac{v_0^2}{\omega^2}} \quad \tan \phi = -\frac{v_0}{\omega x_0} \text{ 检验 } \phi \text{ 一个解}$$

3. 简谐振 - 单摆 $\omega = \sqrt{\frac{g}{L}} \quad \frac{d^2\theta}{dt^2} + \omega^2 \theta = 0 \rightarrow$ 谐振动方程

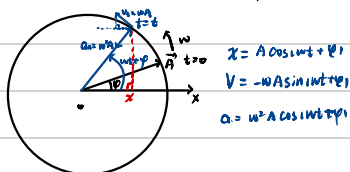
$$\theta = \theta_m \cos(\omega t + \phi)$$

$$\begin{cases} \text{圆频率 } \omega = \sqrt{\frac{g}{L}} \\ \text{角振幅 } \theta_m = \sqrt{\theta_0^2 + \frac{v_0^2}{\omega^2}} \\ \text{初相位 } \phi = \arctan\left(-\frac{v_0}{\omega \theta_0}\right) \end{cases}$$

4. 谐振动的矢量表示法 —— 旋转矢量法

1) 旋转矢量与位移 x

设 $t=0$ 时: $\vec{A}$ 与 x 轴的夹角为 φ



2) 位相 $\omega t + \varphi = (\frac{2\pi}{T}t + \varphi)$

同相差, $x_1 = A_1 \cos(\omega t + \varphi_1)$ $x_2 = A_2 \cos(\omega t + \varphi_2)$

位相差 $\Delta\varphi = \varphi_2 - \varphi_1 = \varphi_2 - \varphi_1$

$\Delta\varphi = \begin{cases} 2k\pi & \text{同相} \\ (2k+1)\pi & \text{反相} \\ > 0 & x_2 \text{ 超前 } \Delta\varphi \text{ 或 } x_1 \text{ 落后 } \Delta\varphi \end{cases}$

5. 机械振动系统的能量

1) 简谐振动系统的动能和势能

水平弹簧振子的能量

$$E_{\text{总}} = E_K + E_P = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

任意时刻: 动能 $E_K = \frac{1}{2}mv^2 = \frac{1}{2}kA^2 \sin^2(\omega t + \varphi)$

势能 $E_P = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \cos^2(\omega t + \varphi)$

$$\omega^2 = \frac{k}{m} \Rightarrow E = E_K + E_P = \frac{1}{2}kA^2$$

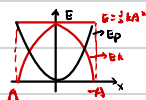
2) 谐振系统能量的特点

$$E_K = \frac{1}{2}kA^2 \sin^2(\omega t + \varphi) \quad E_P = \frac{1}{2}kA^2 \cos^2(\omega t + \varphi)$$

$E_{\text{总}} = \text{常量}$

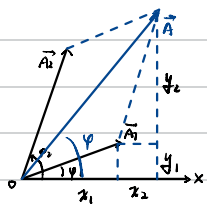
$$\overline{E_K} = \frac{1}{4}kA^2 \quad \overline{E_P} = \frac{1}{4}kA^2 \quad \overline{E_{\text{总}}} = \overline{E_K} = \overline{E_P}$$

$$E_K = \frac{1}{2}k(A^2 - x^2) \quad E_P = \frac{1}{2}kx^2$$



6. 振动的合成与分解

11) 同振动方向, 同频率的两个谐振动合成



$$x_1 = A_1 \cos(\omega t + \varphi_1), \quad x_2 = A_2 \cos(\omega t + \varphi_2)$$

两谐振动的合振动 (仍是同频率的振动) $x = A \cos(\omega t + \varphi)$

$$A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos(\varphi_2 - \varphi_1)$$

$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$

① 两振动同相 $\varphi_2 - \varphi_1 = 2k\pi \quad (k=0, \pm 1, \pm 2, \dots)$

$$A = A_1 + A_2 \quad \varphi = \varphi_1 = \varphi_2 \quad x = (A_1 + A_2) \cos(\omega t + \varphi_1)$$

② 两合振动反相 $\varphi_2 - \varphi_1 = (2k+1)\pi$

$$A = |A_1 - A_2| \quad \text{若 } A_1 > A_2, \varphi = \varphi_1 \quad \text{若 } A_1 < A_2, \varphi = \varphi_2$$

③ 两分振动 $\varphi_2 - \varphi_1 = k\pi$

$$|A_1 - A_2| < A < A_1 + A_2$$

12) 同振动方向, 不同频率的两个谐振动的合成

$$x_1 = A_0 \cos(\omega_1 t + \varphi_1) \quad x_2 = A_0 \cos(\omega_2 t + \varphi_2)$$

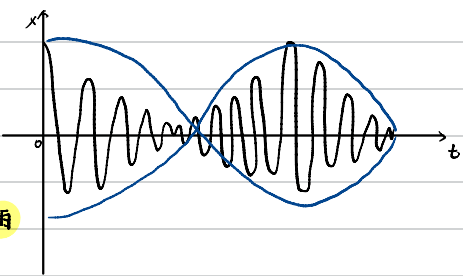
$$x = x_1 + x_2 = A_0 [\cos(\omega_1 t + \varphi_1) + \cos(\omega_2 t + \varphi_2)]$$

$$= 2A_0 \cos \frac{\omega_1 + \omega_2}{2} t + \frac{\varphi_1 + \varphi_2}{2} \cos \frac{\omega_2 - \omega_1}{2} t + \frac{\varphi_2 - \varphi_1}{2}$$

$$x = A(t) \cos(\omega t + \varphi) \quad 0 \leq A(t) \leq 2A_0$$

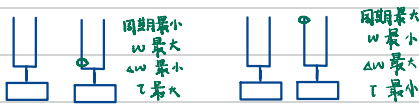
不为谐振动, 振动曲线取决于频率差

若频率差很小, 振幅将出现明显的加强和减弱现象——拍



$$\text{拍的周期 } T = \frac{\omega_1 - \omega_2}{2} = \pi \Rightarrow T = \frac{2\pi}{\omega_1 - \omega_2}$$

$$\text{拍的频率 } \nu = \frac{1}{T} = \frac{\omega_1 - \omega_2}{2\pi} = \nu_1 - \nu_2 \quad \Delta \nu \uparrow \Rightarrow T \uparrow \quad \text{拍现象只在两分振动的频率相差大时才明显}$$



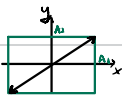
13) 振动方向相互垂直、频率相等的两个谐振动的合成

$$x = A_1 \cos(\omega t + \varphi_1) \quad y = A_2 \cos(\omega t + \varphi_2)$$

$$\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} - \frac{2xy}{A_1A_2} \cos(\varphi_2 - \varphi_1) = \sin^2(\varphi_2 - \varphi_1)$$

① $\Delta \varphi = \varphi_2 - \varphi_1 = 0 \quad \frac{x}{A_1} - \frac{y}{A_2} = 0$

$$y = \frac{A_2}{A_1} x, \text{ 频率: 拍 } \theta = \frac{A_2}{A_1} \quad \varphi_2 = \varphi_1 = \varphi$$



③ $\Delta\varphi = \varphi_2 - \varphi_1 = \pi$, 有 $\left(\frac{x}{A_1}\right)^2 + \left(\frac{y}{A_2}\right)^2 = 0$
 $y = -\frac{A_2}{A_1}x$

④ $\Delta\varphi = \varphi_2 - \varphi_1 = \frac{\pi}{2}$, 有 $\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1$ 

$\varphi_2 = \varphi_1 + \frac{\pi}{2}$ $x = A_1 \cos(\omega t + \varphi_1)$

$y = A_2 \cos(\omega t + \varphi_1 + \frac{\pi}{2}) = -A_2 \sin(\omega t + \varphi_1)$

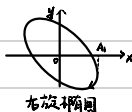
当 $\omega t + \varphi_1 = 0$ 时, $x = A_1, y = 0$

而 $\omega t + \varphi_1 + \frac{\pi}{2}$ 时 $x > 0, y < 0$ $\Rightarrow$ 振动为顺时针方向

⑤ $\Delta\varphi = \varphi_2 - \varphi_1 = \frac{3\pi}{2}$, 同理 $\frac{x^2}{A_1^2} + \frac{y^2}{A_2^2} = 1$

⑥ $\Delta\varphi = \varphi_2 - \varphi_1 = \varphi$, φ 为其它任意值

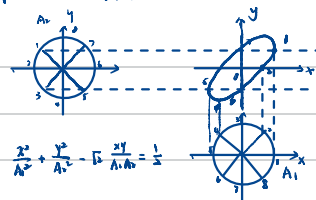
当 $0 < \varphi < \pi, t=0$ 时:



当 $\pi < \varphi < 2\pi$, 或 $-\pi < \varphi < 0$ 时



Ex: 已知两分振动为: $x = A_1 \cos \omega t, y = A_2 \cos(\omega t + \frac{\pi}{2}) \Rightarrow \Delta\varphi = \frac{\pi}{2}$
 求合成振动的轨迹



(2) 若已知 A, ω, φ, m , 求质点在任一位置所受合力

$\vec{F} = m \vec{a} = m \left(\frac{d^2x}{dt^2} \vec{i} + \frac{d^2y}{dt^2} \vec{j} \right)$

$= -m\omega^2 \{ A_1 \cos \omega t \vec{i} + A_2 \cos(\omega t + \frac{\pi}{2}) \vec{j} \}$

$= -m\omega^2 (x \vec{i} + y \vec{j})$

$= -m\omega^2 \vec{r}$

(4) 不同频率、垂直方向谐振动的合成

$x = A_1 \cos(\omega_1 t + \varphi_1) \quad y = A_2 \cos(\omega_2 t + \varphi_2)$

切点数为 $\frac{N_y}{N_x} = \frac{T_1}{T_2} = \frac{\omega_2}{\omega_1}$

7. 谐振动的阻尼振动

1) $F = F_{弹} + F_{阻} = -kx - r \frac{dx}{dt} \quad m \frac{d^2x}{dt^2} = -kx - r \frac{dx}{dt}$

即 $\frac{d^2x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = 0$ 其中 $\omega_0 = \sqrt{\frac{k}{m}}, \beta = \frac{r}{2m}$

2) ① 阻尼较小时, $\beta < \omega_0$, 称为弱阻尼

$x(t) = A_0 e^{-\beta t} \cos(\omega t + \varphi) \quad A = A_0 e^{-\beta t} \quad \omega = \sqrt{\omega_0^2 - \beta^2}$

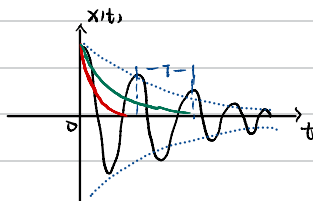
振幅随时间按指数衰减 $\frac{A_0 e^{-\beta t}}{A_0 e^{-\beta(t+\tau)}} = e^{-\beta \tau}$

准周期运动, 准周期为相邻两极大值的时间 $T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\omega_0^2 - \beta^2}}$

阻尼后运动周期比系统固有周期要长 即阻尼延缓了振动, β 越大, 振幅衰减得越快

② 阻尼过大, $\beta > \omega_0$, 称为过阻尼

非周期运动



③ 临界阻尼, $\beta = \omega_0$

11)

8. 受迫振动: 系统在周期性外力下的振动

$$\frac{d^2x}{dt^2} + 2\beta \frac{dx}{dt} + \omega_0^2 x = f_0 \cos \omega t \quad \xrightarrow{\text{稳态解}} \quad x(t) = A_p \cos(\omega t + \alpha)$$

稳态时的受迫振动由驱动力决定。 $F(t) = F_0 \cos \omega t$ $f_0 = \frac{F_0}{m}$

稳态频率 $\omega = \omega_{\text{外}}$

将稳态解代入方程

$$\begin{cases} \text{振幅 } A_p = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2 \omega^2}} \\ \text{相位 } \alpha = -\frac{2\beta \omega}{\omega_0^2 - \omega^2} \end{cases}$$

12) 共振: 在一定条件下, 振幅出现极大值, 振动剧烈的现象.

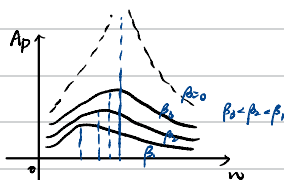
$$A_p = \frac{f_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2 \omega^2}}$$

令 $\frac{dA_p}{d\omega} = 0$ 则 $\omega_r = \omega_0 = \sqrt{\omega_0^2 - 2\beta^2}$ $A_p = A_{\text{max}} = \frac{f_0}{2\beta \omega_r \beta}$

$\omega_r < \omega_0$ 与 β 有关

β 大, ω_r 小, A_{max} 小

若 $\beta \ll \omega_0$ 则 $\omega_r \approx \omega_0$ $A_r = \frac{f_0}{2\beta \omega_0}$ - 称尖锐共振 $\beta \rightarrow 0, A_r \rightarrow \infty, \alpha_r = -\frac{\pi}{2}$



受迫振动相位落后于驱动力 $\frac{\pi}{2}$, 振动速度与驱动力同位相, 外力始终做正功

9. 机械波 - 横波: 振动方向与传播方向垂直 Eg: 电磁波

纵波: 振动的方向与传播方向相同

混合波

11) 产生条件

波源 + 介质 $\xrightarrow{\text{弹性作用}}$ 机械波产生

12) 基本概念

① 波长 λ : 两相邻位相差相差 2π 的质点距离

② 周期 T : 波向前传播一个波长所用的时间 $T = \frac{\lambda}{u} = \frac{2\pi}{\omega} = T_{\text{振}}$

③ 频率 ν : $\nu = \frac{1}{T}$

④ 波速 u : $u = \frac{\lambda}{T}$ $u_{\text{波}} \neq u_{\text{振}}$ 决定于媒质的性质

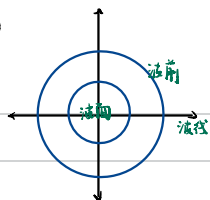
固体 $\begin{cases} u = \sqrt{\frac{G}{\rho}} & \text{切变弹性模量} \\ u = \sqrt{\frac{E}{\rho}} & \text{杨氏模量} \end{cases}$ ρ 为介质密度

流体 $u = \sqrt{\frac{\gamma}{\rho}}$ 体变弹性模量

⑤ 波数 k : 在波的传播方向 λ 内, 长度内包含的波长的个数 $k = \frac{2\pi}{\lambda}$

单位长度内包含的波长个数 $k = \frac{1}{\lambda}$

⑫



12) 平面简谐波

① 波函数:

以横波为例: 设一简谐波以速度 u 向 x 轴方向传播

在坐标原点 O 处引起的振动为: $y_0 = A \cos \omega t + \varphi$

任选一点 P , $OP = x$

注: 波传播的是质点的振动状态, 即波速 = 位相传播的速度 (相速)

O 点位相经 at 传到 P 点, $at = \frac{x}{u}$

则 P 点 t 时刻位相 = O 点 $(t - \frac{x}{u})$ 时刻位相 = $\omega(t - \frac{x}{u}) + \varphi$

任意点 P 振动: $y = A \cos [\omega(t - \frac{x}{u}) + \varphi]$

1° 当 $x = x_1$ 为常数时 $y = A \cos [\omega t + (\varphi - \frac{\omega x_1}{u})] = f(t)$ 表示 x_1 处质点 P 随 t 的振动方程

2° 当 $t = t_1$ 为常数时, $y = A \cos [\omega(t_1 - \frac{x}{u}) + \varphi] = f(x)$ 表示 t_1 时刻沿波线上所有质点的振动状态 —— 介质波动状态。

3° 当 t 为常数, C 为常数时 $x = x_1, t = t_1$

若 $t_2 = t_1 + \Delta t$ 为常数 $x_2 = x_1 + \Delta x = x_1 + u \Delta t$ $y_2 = A \cos [\omega(t_2 - \frac{x_2}{u}) + \varphi] = y_1$

经 Δt 后, 整个波形向前移动了 $\Delta x = u \Delta t$

② 平面波的波动方程

$$\frac{\partial^2 y}{\partial x^2} = \frac{1}{u^2} \frac{\partial^2 y}{\partial t^2} \quad u = \sqrt{\frac{F}{\rho}} \quad \text{与媒质的弹性和惯性有关}$$

13) 波的能量及传播 机械波的能量是媒质中各质元振动能量(振动动能、形变势能)的总和

$$y(x, t) = A \cos [\omega(t - \frac{x}{u})]$$

$$v = \frac{\partial y}{\partial t} = -A\omega \sin [\omega(t - \frac{x}{u})]$$

$$W_K = \frac{1}{2} \rho A^2 \omega^2 \Delta V \sin^2 [\omega(t - \frac{x}{u})] \quad \text{单位长度动能}$$

应力 = 横截面积单位面积上的张力

$$\frac{F}{S} = -Y \frac{\partial y}{\partial x} \quad F = k \Delta y \quad k = Y \frac{S}{\Delta x}$$

$$W_P = \Delta W = \int_0^{\Delta y} k \Delta y = \frac{1}{2} k (\Delta y)^2 = \frac{1}{2} \frac{YS}{\Delta x} (\frac{\partial y}{\partial x})^2 = \frac{1}{2} Y S \Delta V (\frac{\partial y}{\partial x})^2$$

$$W_P = W_K = \frac{1}{2} \rho A^2 \omega^2 \Delta V \sin^2 [\omega(t - \frac{x}{u})]$$

$$\Delta W = W_P + W_K = \rho A^2 \omega^2 \Delta V \sin^2 [\omega(t - \frac{x}{u})]$$

1° W_{Pmax} 即 W_{Kmax} , 如平衡位置

2° W_{Pmin} 即 W_{Kmin} , 如最大位移处

介质中质元 不守恒, 能量传播
(守恒)